

Credit Card Fraud Detection with XAI: Improving Interpretability and Trust

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Abstract—Fraud detection is crucial for financial institutions to protect clients' funds and maintain trust in online payment systems. However, traditional machine learning models often lack interpretability due to advanced fraud methods. This study proposes a framework that integrates advanced machine learning algorithms with XAI to improve the interpretability of fraud detection models. The framework explains model predictions clearly, helping stakeholders understand and verify system decisions. The Kaggle dataset is pre-processed, i.e., classes are balanced before implementing the ML techniques. The pre-processed Kaggle dataset is used to implement various ML algorithms, with the Local Interpretable Model-agnostic Explanations (LIME) method used during the prediction phase. Decision tree and Random Forest outperform other ML techniques, achieving 100% accuracy. This method clarifies the decision-making process, allowing stakeholders like financial analysts, lawmakers, and consumers to trust and confirm system results.

Keywords—artificial intelligence, machine learning, LIME, finance, tree

I. INTRODUCTION

Many financial organizations now offer online banking services to the public. Electronic payment systems are crucial in today's competitive financial environment. They have significantly improved the ease of buying goods and services. Financial institutions often provide cards to clients for convenient electronic transactions. Customers must verify transactions with the retailer before finalizing any credit card transaction [1].

Credit cards offer numerous advantages to users, but they are also linked to issues like security and fraud. Banks and financial organizations are now grappling with the challenge of credit card fraud. It happens when unauthorized people rack up debt by fraudulently charging goods and services on their credit cards. Fraudulent actions linked to credit and debit cards significantly contribute to financial losses in the finance business [2]. The rapid technological advancement poses a significant hazard, resulting in substantial financial losses worldwide. Because of this, credit card fraud monitoring is necessary to keep money from being lost.

The utilization of machine learning (ML) and artificial intelligence (AI) in credit card fraud detection is driven by the need for more effective, precise, and adaptable systems to counter the development of fraudulent methods. Fraudsters are constantly adjusting their strategies to evade conventional rule-based fraud detection systems. These systems frequently have difficulty keeping up with the constantly changing fraudulent activities. Artificial intelligence (AI) and machine learning (ML) methods provide the capacity to analyze extensive transaction data, identify intricate patterns, and instantly adjust to emerging fraud trends [3]. Adaptability is

essential for outsmarting fraudsters who continuously create new tactics to exploit weaknesses.

Manual detection approaches are inefficient and error-prone due to their intricate nature and the vast amount of financial data. Due to the high volume of daily transactions, it is hard for humans to review each one thoroughly. AI and ML algorithms are proficient at handling and examining extensive datasets, revealing complex connections and irregularities that could suggest fraudulent activity. The algorithms can efficiently analyze large volumes of data, detecting patterns that suggest fraudulent transactions with great precision [4].

Real-time fraud detection is crucial for preventing unlawful transactions and reducing financial losses. Conventional methods often rely on batch processing, leading to delays. AI and ML enable instantaneous fraud detection by continuously evaluating incoming transactions, identifying suspicious behaviors, and prompting alarms for quick inquiry [5]. Adaptive and self-learning systems use supervised, unsupervised, and reinforcement learning methods to improve performance and differentiate between genuine and deceptive transactions. This adaptability is essential in the ever-evolving realm of financial fraud. Efficient fraud detection secures financial institutions and protects clients from unlawful access to their accounts, reducing discomfort, financial loss, and erosion of trust.

Despite their immense potential in identifying fraudulent transactions, machine learning algorithms are frequently perceived as "black boxes" because of their complexity. By offering insights into these algorithms' decision-making processes, Explainable Artificial Intelligence (XAI) approaches seek to overcome this problem and improve the reliability and usefulness of these algorithms in crucial applications like credit card fraud detection [6]. It can assist in identifying errors and enhancing models. Analysts can detect any biases or flaws in the model's decision-making process by comprehending the explanations offered by XAI approaches. If the model repeatedly identifies fraudulent transactions from a particular demographic group, it may suggest bias in the data or model.

Analysts can use these observations to enhance the model, boost its efficiency, and minimize false positives and negatives. Iteratively improving the model is essential for ensuring the efficacy of fraud detection systems [7]. Aside from fraud detection, AI and ML can be applied to a variety of disciplines, including computer vision [8], customer segmentation [9], cloud computing [10], data quality [11], plausible applications [12], image processing [13], and water quality prediction [14].

This research significantly contributes by creating an innovative XAI framework for detecting fraudulent credit card activity. The pre-processed data is subjected to several

machine learning (ML) techniques using the suggested framework, including Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), K closest neighbor (KNN), and Naive Bayes (NB). The Synthetic Minority Over-sampling Technique (SMOTE) technique balances the dataset. In conclusion, the XAI approach is utilized to analyze the outcomes that were acquired by the trained models in order to incorporate trustworthiness into the proposed framework.

II. LITERATURE REVIEW

Various researchers have proposed ML frameworks for credit card fraud detection. Ileberi et al. [15] suggest a credit card fraud detection system based on machine learning (ML) that utilizes the genetic algorithm (GA) to identify features. Following the selection of the features that have been optimized, the detection engine employs various ML techniques like DT, RF, NB, Artificial Neural Networks (ANN), and LR. The performance is validated using a dataset created from European cardholders.

Ileberi et al. [16] also utilize an ML to detect credit card fraud by analyzing skewed datasets from European credit cards. The class imbalance issue was resolved by re-sampling the dataset using SMOTE. The framework was assessed using the ML methods: Support Vector Machine (SVM), LR, DT, RF, Extreme Gradient Boosting (XGBoost), and Extra Tree (ET). The ML algorithms were enhanced using the Adaptive Boosting (AdaBoost) technique to improve their classification accuracy. The highest accuracy achieved is 99.98% using Adaboost and Extra Tree.

Similarly, Saheed et al. [17] developed a novel model utilizing principal component analysis (PCA) for feature selection and supervised machine learning algorithms, including KNN, ridge classifier, gradient boosting, quadratic discriminant analysis, AdaBoost, and RF for distinguishing between fraudulent and legitimate transactions. The KNN model performed best on the German data set with accuracy, recall, and precision of 96.29%, 100%, and 96.29%, respectively, while on the Taiwan dataset, the ridge classifier outperforms with accuracy, recall, and precision of 81.75%, 34.89, and 66.61%, respectively.

Likewise, Madhurya et al. [18] analyzed the anomaly of class imbalance, and strategies for its implementation are examined to provide specific outcomes in credit card fraud detection. The efficacy of the algorithms fluctuates depending on the dataset and the specific scenario in which they are applied. It is demonstrated that all algorithms exhibit a specific imbalance at some time during the investigation, regardless of the calculations. The study revealed that logistic regression had higher accuracy, but learning curves indicated that most of the algorithms underfit, whereas KNN could only learn. Therefore, KNN is a superior classifier, achieving an accuracy of 95.89%.

John Unogwu & Filali [19] also investigate various ML algorithms for discerning legitimate or fraudulent transactions using the Credit Card Fraud dataset. In order to address the class distribution imbalance caused by the dataset's high imbalance, the SMOTE technique was employed for oversampling. Furthermore, feature selection was performed, and training and test data were generated from the dataset. The experiments employed NB, RF, and MLP algorithms. The MLP approach attained a 99.95% accuracy rate compared to alternative methods. Jain et al. [20] apply ML algorithms to a dataset of credit card frauds and compare the effectiveness of

three different algorithms in detecting fraudulent credit card activities. The study proved that the RF method is more accurate than the DT and XGBOOST algorithms.

Alternatively, some researchers have used deep learning (DL) methods like Mienye & Sun [21] suggest a robust DL method that utilizes long short-term memory (LSTM) and gated recurrent unit (GRU) in a stacking ensemble setup and Multilayer perceptron. The SMOTE-ENN method, a hybrid synthetic minority oversampling technique combined with an edited nearest neighbor, is used for data balancing. The suggested model achieves a sensitivity of 1.000 and a specificity of 0.997.

Dang et al. [22] examined and assessed the latest developments in ML algorithms and deep reinforcement learning applied to CCF detection systems, which involve identifying fraud and non-fraud instances. SMOTE and ADASYN resampling techniques are applied to rebalance the skewed CCF dataset. Machine learning algorithms are then used on this balanced dataset to create CCF detection systems. Subsequently, DRL is utilized to develop detection systems using the imbalanced CCF dataset. Different categorization metrics are used to comprehensively assess the performance of machine learning and deep reinforcement learning models. Applying SMOTE and ADASYN to resample the original CCF datasets before splitting them into training and test sets results in machine learning models, achieving accuracy rates of over 99%.

Like the above approach, Alharfaj et al. [23] have performed an empirical investigation utilizing the European card benchmark dataset for fraud detection. The first application of an ML algorithm to the dataset slightly enhanced fraud detection accuracy. Three convolutional neural networks (CNN) designs are later implemented to enhance fraud detection performance. Adding more layers significantly improved the accuracy of detection. In-depth research was done by modifying the number of hidden layers and epochs. The study's evaluation reveals better results in accuracy (99.9%), f1-score (85.71%), precision (93%), and area under the curve (98%) than before.

According to the literature study, ML approaches such as RF, DT, KNN, MLP, ANN, NB, and Boosting are widely recognized and show promise for detecting credit card fraud. Recurrent neural networks, such as LSTM, CNN, etc., are commonly used in deep learning and are known for producing favorable outcomes. Preprocessing techniques are applied to all datasets to balance them and improve accuracy. This study will utilize fundamental pre-processing methods and SMOTE to balance the data. The ML technique is then utilized on the pre-processed data. After training the models, performance evaluation is conducted using the test data. The test results are validated using the XAI technique to assess the reliability of the proposed system.

III. METHODOLOGY

This section describes the proposed method in detail. The paper proposed an XAI ML-based credit card detection (XMCCD) framework. The proposed diagram in Fig 1 provides the visual details of the proposed method.

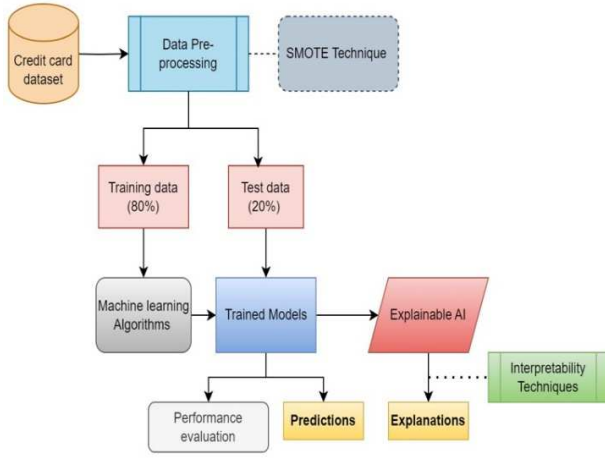


Fig. 1. Proposed methodology for XAI ML-based credit card detection

A. Dataset description

Credit card fraud detection, an open-source dataset from Kaggle [24], is utilized for the experiments. The dataset includes credit card purchases made by European consumers in September 2013. The dataset contains all the information about the activities that took place for two days. Among a total of 284,807 transactions, 492 were found to be fraudulent. There is a substantial imbalance in the dataset, as just 0.172% of the events fit into the positive category, which is fraudulent. The dataset is made up of only numerical input factors that were created using a method called Principal Component Analysis (PCA). Table 1 contains the specifics of the characteristics included in the dataset.

TABLE I. DATASET DESCRIPTION

Features	Description
V1, V2, ... V28	PCA derived components
Time	Time in seconds that elapses between every transaction and the dataset's first transaction
Amount	The amount of money involved
Class	When fraud occurs, the value is 1; otherwise, it is 0

B. Pre-processing

The various pre-processing steps, like identifying missing values and duplicates, are performed. The dataset contains no missing values. However, there were 2081 duplicate rows in the dataset. After removing them, the dataset consists of 283726 Rows and 31 columns. The dataset is highly imbalanced, consisting of only 492 Fraud instances. Thus, the Synthetic Minority Over-sampling Technique (SMOTE) technique is used to balance the classes in the dataset.

SMOTE [25] is an ML technique designed to address the class imbalance in the dataset where a minority class is less represented than the dominant class. Fraudulent transactions are often less familiar than lawful ones in fraud detection, making them the minority class. SMOTE creates artificial instances of the minority class to equalize the class distribution in the dataset. The process begins by recognizing the minority class in the dataset, such as the category for illegitimate transactions. After identifying examples, SMOTE creates artificial samples by interpolating between these instances and their closest neighbors in the feature space. The procedure iterates until the appropriate balance between minority and majority classes is reached or a specific number of synthetic samples is created. However, SMOTE's effectiveness depends on fine-tuning the parameters, such as

the number of closest neighbors and the potential for overlap or noise in the dataset. Therefore, assessing the model's performance using different validation or test data to verify its effectiveness in real-world situations is essential.

After applying SMOTE, the number of samples in the dataset is 453,194, 22,6597 for each class.

C. Machine learning models

Once the synthetic samples are created using SMOTE, the dataset is divided into training and testing, with a ratio of 4:1. The number of training samples is 226980 and 30 features, while for the test data, there are 56746 samples. The training data is provided for five different ML techniques: LR, DT, RF, NB, and KNN [26].

1) *Logistic Regression*: (LR) is a classification technique used for binary categorization, estimating the likelihood of a certain input belonging to a certain class. It is effective when the connection between characteristics and the target variable is linear or can be made linear using feature engineering.

2) *Decision Trees*: (DT) are flexible supervised learning algorithms used for classification and regression applications, recursively dividing the feature space. They are user-friendly and straightforward but susceptible to overfitting. Random Forests (RF) is an ensemble learning approach that extends decision trees, enhancing generalization and mitigating overfitting.

3) *Naive Bayes*: (NB) is a probabilistic classification technique that applies Bayes' theorem under the feature independence assumption. It is computationally efficient and requires minimal training data, making it suitable for large-scale applications.

4) *K-nearest neighbors*: (KNN) is a non-parametric classification approach that predicts based on the majority class or mean value of the k-nearest neighbors of a data point. However, its efficiency may diminish when dealing with high-dimensional and large datasets due to computational complexity.

D. Explainable AI

Once the models are trained, the test data is given to make predictions on the trained ML models. The predicted results are interpreted using an interpretable technique like Local Interpretable Model-agnostic Explanations (LIME) [27]. LIME is a method within XAI that aims to offer understandable explanations for specific predictions generated by intricate machine learning models. It deals with the opaque characteristics of complex models by estimating their behavior in a limited area surrounding particular cases.

LIME is an explainable artificial intelligence (XAI) technique designed to clarify the decision-making process of intricate machine learning models on a per-instance basis. It achieves this by developing understandable substitute models, like linear models or decision trees, centered on a particular prediction of interest. LIME alters the input data close to the specific instance and analyzes the impact on predictions to create a localized replica of the model's parameters. To understand how specific features impact the model's decision-making, LIME examines the substitute model's parameters or the significance of features. Because of this, users can acquire a better grasp of the reasons behind a given prediction, which

ultimately increases confidence, transparency, and accountability in artificial intelligence systems.

E. Performance measures

The standard performance metrics used to evaluate the proposed framework are below [28].

Accuracy is a measure that calculates the proportion of adequately identified cases out of the total number of instances. The accuracy equation is provided in Equation 1.

$$Accuracy = \frac{\text{Number of samples correctly classified}}{\text{Total number of samples}} \quad (1)$$

Precision is a measure that calculates the ratio of correctly predicted positive outcomes to all the projected positive outcomes. The main focus is on providing accurate and optimistic predictions, as given in Equation 2.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

Recall is the percentage of correctly predicted positive outcomes compared to the total number of true positives in the dataset. The primary goal is to record all favorable occurrences as given in Equation 3.

$$Recall = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

The F1 score provides a balanced compromise between precision and recall, especially when dealing with imbalanced class distributions. The F1 score is defined in Equation 4.

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The Receiver Operating Characteristics (ROC) curve is a visual tool utilized to evaluate the effectiveness of binary classification algorithms. The graph displays the sensitivity vs. 1-specificity at different threshold settings, demonstrating the balance between accurately identifying positive and mistakenly identifying negative cases. An ideal classifier would provide an ROC curve intersecting the point (0,1), denoting flawless classification with zero false positives. Conversely, random guessing would produce a ROC curve following the diagonal line extending from (0,0) to (1,1). The Area under the curve (AUC)-ROC is a summary statistic that measures model performance, where larger values suggest stronger discriminating ability. The ROC curve offers a succinct visual depiction of a classifier's capacity to differentiate between positive and negative examples, aiding in model assessment and comparison.

IV. RESULTS AND ANALYSIS

Experiments are performed using Python and the P100 GPU on Google Colab. The results obtained after testing the trained models on the test data are described in Table II; the comparison of all the models is given in Fig 2, and the comparison of ROC curves for all the models is in Fig. 3.

TABLE II. TESTING RESULTS FOR PREDICTING CREDIT CARD FRAUD

Models	Accuracy	Precision	Recall	F1 score	AUC	Time (secs)
LR	97 %	97 %	97 %	97 %	0.99	7.96
DT	100 %	100 %	100 %	100 %	0.99	59.79

RF	100 %	100 %	100 %	100 %	0.99	572.9
KNN	96 %	96 %	96 %	96 %	0.98	98.71
NB	88 %	90 %	88 %	88 %	0.98	0.46

Table II and Fig 2. demonstrate that the DT and RF models attain a 100% accuracy rate with an AUC of 0.99. The LR model achieved the second-greatest accuracy of 97%, followed by KNN with 96% accuracy, while NB had the lowest performance with 88% accuracy. Attaining 100% accuracy is feasible due to the dataset being balanced and duplicate rows being eliminated before model training. This paper does not address the outliers. Therefore, models such as Naive Bayes (NB) and K-Nearest Neighbors (KNN) exhibited poor performance due to outliers in the dataset. Tree-based models effectively manage outliers, resulting in a 100% accuracy rate for these models. The graph in Fig 3 shows that the top-performing models are DT and RF since they intersect at the location (0,1), which is shown in blue.

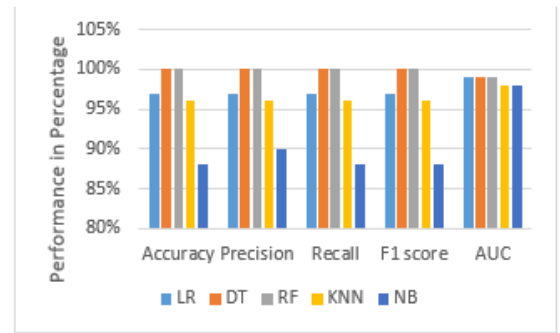


Fig. 2. Comparison of all the ML models with standard performance measures

The predictions made by the model are analyzed using the LIME, the XAI technique, to interpret the results obtained from the models. The interpretation of the prediction made by the DT model for the feature having the feature value for the first ten features is shown in Fig 3.

The decision tree selected the top 10 characteristics as V17, V10, V12, V3, V4, V7, V16, V8, V24, and V11, as shown in Fig 4. The forecast is based on the top 10 features. Blue features indicate no fraud when the feature value meets the specified criteria in Fig. 4, whereas orange features indicate fraud when the feature value meets the criterion. Feature V17 will be classified as No fraud if its value exceeds 0.18. The feature value of 0.23 suggests that there is no fraud. For feature V4, the value should be more than -0.06 and less than or equal to 1.49. The feature value of V4 for the prediction is 1.12, indicating fraud. Out of the ten features, 7 meet the criteria for no fraud, resulting in a probability of 1 for no fraud.

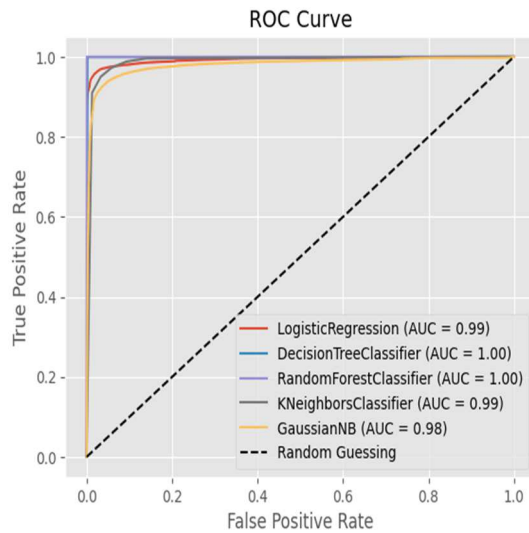


Fig. 3. Comparison of all the ML models with ROC curve

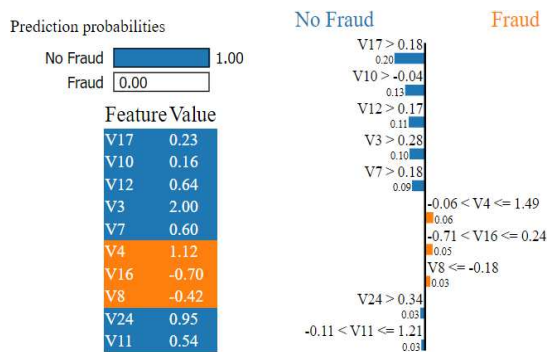


Fig. 4. Interpretation of results for DT model using LIME technique

V. CONCLUSION

This paper enhances credit card fraud detection by balancing the complexity of models and understanding. Integrating cutting-edge machine learning algorithms and XAI methodologies effectively illustrates how interested parties can acquire significant knowledge regarding the determinants that impact fraud forecasts. The proposed framework enables users to grasp and authenticate the system's results by providing human-comprehensible justifications for model choices, thus cultivating confidence and trust in its functionalities. The proposed system uses the Kaggle dataset and implements ML techniques like LR, DT, RF, KNN, and NB after balancing the classes in the dataset using the SMOTE technique. Decision trees and Random Forests are the top-performing ML approaches, with 100% accuracy. The XAI technique, the LIME, is utilized to explain the model decisions and to understand which features play an essential role in predicting the test data. Financial organizations may use the proposed framework to enhance security and maintain trust in the digital payment environment by implementing XAI to create less opaque and reliable systems for identifying fraud. Despite achieving the highest performance, the research can be extended to implement the proposed framework on real-time datasets to analyze its efficiency.

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